

Multiple Choice (10 marks)

1. To verify a digital signature we need the
 - (a) Sender's Private key
 - (b) Sender's Public key
 - (c) Receiver's Private key
 - (d) Receiver's Public key
2. In a _____ attack, the extra data that holds some specific instructions in the memory for actions is projected by a cyber-criminal or penetration tester to crack the system.
 - (a) Phishing
 - (b) MiTM
 - (c) Buffer-overflow
 - (d) Clickjacking
3. The _____ is categorized as an unknown segment of the Deep Web which has been purposely kept hidden & is inaccessible using standard web browsers.
 - (a) Haunted web
 - (b) World Wide Web
 - (c) Dark web
 - (d) Surface web
4. Which of the following does authorization aim to accomplish?
 - (a) Restrict what operations/data the user can access
 - (b) Determine if the user is an attacker
 - (c) Flag the user if he/she misbehaves
 - (d) Determine who the user is
5. _____ is the anticipation of unauthorized access or break to computers or data by means of wireless networks.
 - (a) Wireless access
 - (b) Wireless security
 - (c) Wired Security
 - (d) Wired device apps

True / False (10 marks)

Answer True or False

6. True or False: Memory leakage is a critical vulnerability that directly leads to unauthorized system access and operational failures.
7. True or False: The Metasploit framework simplifies the exploitation of vulnerabilities through a user-friendly, point-and-click interface.
8. True or False: It is impossible to derive any bits of the OTP key from the original message and its ciphertext.
9. True or False: Nmap determines port states as open, filtered, and unfiltered during network scanning.
10. True or False: DaCryptic is known as a remote Trojan that primarily targets mobile devices for data theft.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the four-step authentication progression that occurs between a wireless user and an access point during the authentication process, highlighting the significance of the 4-way handshake.
12. Discuss the principles and effectiveness of mutation-based fuzzing in cybersecurity, emphasizing how it generates new inputs through modifications of existing ones.

End of Paper